LESSON PLAN

NBCD (30)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Construct, test and observe output on voice transmission using amplitude modulation and demodulation on optical fiber communication trainer.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Construct, test and observe output on voice transmission using amplitude modulation and demodulation on optical fiber communication trainer.
5. Specific Objectives	30.1.1.1 Trainee should be able to practical on various lab FTL, RF & Embedded lab.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Construction of AM circuit, testing voice transmission, observing demodulated output, safety precautions regarding high voltage and optical sources, instructor supervision during setup and operation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the basic principle of Amplitude Modulation?
(b) How does an optical fiber communication system transmit data?
(c) What are the safety precautions when handling optical fiber components?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to AM Modulation Theory • Components of Optical Fiber Communication Trainer • Circuit Construction and Wiring Procedures • Signal Generation and Input Configuration • Demodulation Process and Output Observation	L & Ds	PPT, IB	15 min
(b)	• Adjustment of Modulation Index for Optimal Transmission • Verification of Voice Signal Integrity • Troubleshooting Common Transmission Errors • Documentation of Observed Waveforms • Comparison of Modulated and Demodulated Signals	L & Ds	PPT, PDF	20 min
(c)	• Final System Performance Evaluation • Recording of Test Results and Observations • Review of Safety Protocols During Operation • Cleanup and Return of Equipment • Preparation for Next Session	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between modulation index and signal quality in AM transmission.
(b) Describe the steps taken to troubleshoot a distorted voice output on the trainer.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between RF and optical fiber transmission methods as discussed in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Operation of maritime radar trainer and measuring the range of the target.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min